MILITARY SPECIFICATION

TRANSMITTING SET, INFRARED AN/UAT-2( )

1. SCOPE

1.1 This specification covers an infrared transmitting set, AN/UAT-2( ), an infrared beacon for use in amphibious operations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Fiberboard
PPP-T-60 - Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive paper, (for Carton Sealing).

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-D-3464 - Dessicants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-B-8565 - Batteries, Storage, Aeronautical
MIL-S-8805 - Switches and Switch Assemblies, Sensitive and Push (Snap Action), General Specification for
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification
MIL-O-16898 - Optical Elements; Packing of.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of
MIL-S-21604 - Switch Rotary, Multipole and Selector Type, 1 to 10 Ampere
MIL-I-45208 - Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-810 - Environmental Test Methods
MS 25306 - Switch, Toggle, Positive Break, Aircraft, Single Pole, One Hole Mounting

PUBLICATIONS

DSAM 4145.3 - Packaging and Handling of Dangerous Materials for Transportation by Military Aircraft; Packaging and Materials Handling
NAVSHIPS 0967-232-7010 - Instruction Manual for Infrared Transmitting Set AN/UAT-2A

FSC 5850

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd., Chicago, Ill. 60606).

NATIONAL CLASSIFICATION BOARD

National Motor Freight Classification Rules

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N.W., Washington, D.C. 20036.)

3. REQUIREMENTS

3.1 First article sample.- Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.2).

3.2 General requirements.- The equipment shall meet the requirements of MIL-E-16400 to the extent specified herein.

3.2.1 Environmental conditions.- The equipment shall be capable of continuous reliable operation under temperature and humidity conditions for class 2 equipment as specified in MIL-E-16400, except that the operating temperature shall be -28° to +60°C. When subjected to the nonoperating conditions for temperature and humidity for class 2 of MIL-E-16400, the equipment shall not be damaged nor its performance degraded.

3.2.2 Mechanical design.- The equipment shall conform to the mechanical design requirements of MIL-E-16400 regarding construction, welding, soldering, mechanical operation, and accessibility.

3.2.2.1 Vibration.- The equipment shall be vibrated in accordance with MIL-E-16400.

3.2.2.2 Inclination.- The requirements for inclination shall be in accordance with MIL-E-16400.

3.2.2.3 Enclosure.- The equipment shall be watertight in accordance with MIL-STD-108. Enclosures shall include stuffing tubes for cable entrances (See 4.5.9).

3.2.3 Electrical design.- The electrical design of the equipment shall conform to the requirements of MIL-E-16400 regarding electrical bonding, internal cabling, connecting wire termination, and identification of conductors.

3.2.3.1 Insulation resistance.- The insulation resistance between all current-carrying and non-current-carrying parts shall be not less than 100 megohms. Resistance shall be measured with an applied potential of 500 volts 60 Hz.

3.2.4 Parts.- Use of parts in the construction of the equipment shall be in accordance with the requirements of MIL-E-16400 regarding design application, interchangeability, gaskets, switches, and threaded devices.

3.2.5 Surplus parts.- Surplus parts manufactured for previous Navy contracts which meet the requirements of this specification may be used provided that:



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(a) The supplier furnishes the Government inspector the Navy Contract number under which the part was fabricated.
(b) All parts are retested and meet the requirements of the current specification covering that part.

3.2.6 Materials.- The materials used in the construction of the equipment shall be in accordance with the materials section of MIL-E-16400 regarding fungus-inert materials, unacceptable materials, arc-resistant materials, tape lacing, metals, plastics, impregnating, filling, potting and encapsulating compounds, and electrical tubing and sleeving.

3.2.6.1 Metals.- Metal parts shall be made of nonferromagnetic corrosion-resistant materials. Ferromagnetic parts may be used only with written authorization of the command or agency concerned

3.2.7 Coating and treatments.- Coatings and treatments shall be in accordance with the coating and treatments provisions of MIL-E-16400.

3.2.8 Thermal design.- Thermal design considerations shall conform to applicable sections of MIL-E-16400.

3.2.9 Controls and indicators.- Use of controls and indicators shall be consistent with requirements of MIL-E-16400.

3.2.10 Supplementary requirements.- Supplementary requirements of MIL-E-16400 regarding serial numbers, date of manufacture, instruction plates, Government nomenclature, and identification and markings shall be complied with unless otherwise specified in the contract or purchase order.

3.2.11 Visual security.- There shall be no white light leaks from any part of the equipment when it is operating.

3.2.12 The equipment shall be able to withstand the drop test specified in 4.5.12 and the salt spray test specified in MIL-E-16400.

3.3 Detailed requirements.-

3.3.1 Functional description.- Except for an external recharging capability, the equipment shall be functionally and operationally equivalent to the AN/UAT-2A infrared Transmitting Set, which is described in NAVSHIPS 097-232-7010. The equipment shall be a pack-type infrared transmitter designed for operation as a steady state beacon or a manually keyed blinker. It shall radiate omni-directionally and uniformly within ±20 degrees of the horizontal plane.

3.3.2 Equipment composition.- The AN/UAT-2( ) infrared transmitting set shall consist of the following:

| | |
|---|---|
| (a) Infrared transmitter (beacon) | 1 each |
| (b) Mast | 1 each |
| (c) Storage battery | 1 each |
| (d) Spare lamps | 1 each |
| (e) Technical manuals | 2 each |

3.3.3 Weights and dimensions.- The general configuration of the equipment is optional within the following limits: the equipment, assembled and ready for use but without the mast extended, shall have no dimension greater than 16 inches and the product of the three major dimensions shall not exceed 870 cubic inches. The overall weight shall not exceed 25 pounds.

3.3.4 Infrared transmitter.- The infrared transmitter shall be designed to be mounted on the battery case cover to which it shall be attached by means of a quick connect-disconnect device. The same device shall be used to attach the transmitter to the mast. The transmitter shall be normally attached to the battery case for transportation and operation. The transmitter shall be watertight in accordance with MIL-STD-108



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to a minimum depth of 3 feet of water. The major components of the transmitter shall consist of a transmitter base fixture, an infrared filter and a lamp.

3.3.4.1 Transmitter base fixture.- The transmitter base fixture shall be made of aluminum or high impact plastic. The base fixture shall be so designed that the infrared filter may be attached to the base by means of a threaded cap ring. The cap ring shall be fluted or grooved on the outer face to provide a nonslip finger grip.

3.3.4.2 Infrared filter.- The infrared filter shall be a molded shape with a closed top and cylindrical side. The open base of the top shall have a flanged rim for attaching it to the base assembly by means of the cap ring. The cylindrical side shall be configured as a Fresnel lens. Height of the filter/globe shall not exceed 6 inches, and the outside diameter exclusive of the flange, not more than 4 inches.

3.3.4.2.1 Filter material.- The material of the filter/globe shall be heat-treated glass or heat- and impact-resistant plastic. Material shall be of good optical quality without strains, seeds or bubbles.

3.3.4.2.2 Filter optical characteristics.- The Fresnel lens section of the filter dome shall be designed to gather and direct as much of the radiation from the beacon lamp as possible into the solid angle specified in 3.3.1. The optical gain of the lens shall be not less than four times.

3.3.4.2.3 Visual and infrared transmission.- The infrared filter shall be in the form either of coloring matter dispersed uniformly throughout the material of the globe or a coating applied to the inner surface. The filter shall not fade, peel or develop pinholes as a result of any of the tests specified herein. The filter shall restrict the visibility of the lamp while transmitting of the maximum of infrared energy. For the 32 cp tungsten filament lamp in accordance with 3.3.4.3 the filter shall have a visual transmission (EVT) not greater than $2 \times 10^{-6}$ and an infrared transmission (EHT) not less than 20 percent in the spectral region from 0.75 to 1.2 microns as measured on an infrared filter photometer utilizing a receiver with an S-1 spectral response (see 4.5.3). Alternatively, measurements may be obtained on a visible-near infrared spectrophotometer and converted to EVT-EHT values.

3.3.4.2.4 Solar radiation.- Visual and infrared transmission characteristics of the filter shall not be degraded by exposure to solar radiation of the intensity normally encountered in an outdoor shipboard environment.

3.3.4.2.5 Mechanical and thermal strength.- The infrared filters shall be properly constructed and processed to assure adequate mechanical and thermal strength. Each filter shall be capable of withstanding the thermal strength test of 4.5.4 and the thermal shock test of 4.5.5 without damage.

3.3.4.3 Infrared transmitter lamp.- The infrared transmitter lamp shall be a Mazda number 1327, or equivalent, automotive spotlight lamp. This is a tungsten lamp with prefocus base. It is rated at 32 candlepower and 2.08 amperes from a 12.75 volts source. The design operating life is 300 hours. A spare lamp shall be provided with the equipment. It shall be secured either within the beacon assembly or the battery case in such a manner that it cannot be shaken loose in transit.

3.3.5 Battery case.- The battery case shall consist of a one-piece box and a cover of aluminum alloy. It shall be designed to receive a 12 volt storage battery. All interior surfaces of the battery case shall be coated with a black, potassium hydroxide-resistant finish. Adequate space shall be provided between the storage battery and surrounding walls so the battery may be easily removed. The battery shall be held immovably in place by either a "hold down" or a resilient liner when the cover is in place. The battery case shall be watertight in accordance with MIL-STD-108 to a minimum depth of 3 feet of water.

3.3.5.1 Battery case cover.- The cover shall be attached to the case by means of captive wing bolts or thumb screws. The carrying handle, switches, and beacon shall be attached to the cover. It shall include a pressure relief valve to vent the interior of the battery case. The carrying handle shall be arranged so that the complete unit may be conveniently carried in the normal vertical position. The hand grip shall be not less than one inch in diameter.



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3.3.6 Switches.-

3.3.6.1 Power switch.- An on-off power switch shall be provided. It shall be a waterproof rotary type switch with detent action conforming to MIL-S-21604, or a toggle switch conforming to MS 25306-311.

3.3.6.2 Keying switch.- A waterproof momentary contact switch shall be provided for manually keying the beacon lamp. Either a separate switch or a momentary contact position on the power switch may be utilized for this purpose. Separate switches shall conform to MIL-S-8805.

3.3.6.3 Switches shall be located on the cover in positions where they are protected in transit and will not be actuated inadvertently.

3.3.7 Mast.- The mast shall be made of aluminum alloy tubing with an anodized protective finish. It shall be capable of supporting the beacon at variable heights between 1 and 5 feet above the battery case. The mast shall be attached to the transmitting set for stowing when it is not in use. The height of the stowed mast shall not exceed 11 inches above the bottom surface of the battery case. The stowed mast shall be attached to the cover in such a way that it will not interfere with the carrying handle or the operation of the beacon. The mast may be either telescoping or sectional as shown in NAVSHIPS 0967-232-7010.

3.3.8 Power cable.- The power cable shall be permanently connected to the beacon and the battery case cover. It shall be a UL approved retactile (helically coiled), neoprene jacketed cable with an extended length of 5-$\frac{1}{2}$ feet.

3.3.9 Storage battery.- The storage battery shall be of the nickel alkaline type. It shall consist of 10 plastic-cased cells connected in series to produce a nominal output of 12 volts. The battery shall have a minimum capacity of 15 ampere hours at the 1-hour discharge rate. It shall power the lamp specified in 3.3.4.3 for 8 hours before the output voltage drops below 10 volts. It shall be capable of recharge from a discharged condition to 90 percent of full charge in 1 hour or less on conventional charging equipment. The temperature operating range for the battery shall be -28° C to +60° C. Cells shall be of spill-proof construction equipped with pressure vents. They shall not leak in any position. Batteries shall conform to the storage, capacity, charge retention, life and cycling test requirements of MIL-B-8565 (see 4.5.11).

3.3.10 Wiring diagram and charging instructions.- Instruction plates containing simplified wiring diagram and charging instructions shall be mounted on the external surface of the battery case (see 3.2.10).

3.3.11 External recharging.- A means of recharging the storage battery without disassembling components or removing the battery case cover shall be incorporated in the design of the transmitting set. A polarized power receptacle shall be mounted on the battery case for external cable connection during recharging. A threaded receptacle cover shall be provided which shall maintain a watertight seal to a minimum depth of 3 feet of water in accordance with MIL-STD-108. The receptacle cover shall be permanently attached to the battery case. A polarized mating plug shall be provided with each equipment supplied.

3.4 Workmanship.- Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system.- The supplier shall provide and maintain an effective inspection and quality assurance system, acceptable to the Government, covering the supplies under the



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contract. The quality assurance system shall be in accordance with the requirements of MIL-I-45208. A current written description of the system shall be submitted to the cognizant Government inspector, for approval (see 6.3). Any changes to the approved plan which might affect the degree of assurance required by this specification or other applicable documents shall be submitted to the cognizant inspector and approved in writing prior to use.

4.1.2 Government verification. - All quality assurance operations performed by the supplier will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all requirements of this specification. Failure of the supplier to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Test, test equipment and test procedures. - The supplier shall submit a list of proposed tests and written test procedures, including test equipment and tolerance limits to be used for first article inspection and quality conformance inspection to Naval Electronic Systems Command, Washington, D. C. 20360, via the Government inspector for review and approval. (See 6.3).

4.2 Classification of inspection. - The method of examination and testing of the equipment fall within the following classifications:

(a) First article inspection
(b) Quality conformance inspection
    (1) Production inspection
    (2) Production control inspection
    (3) Environmental tests

4.3 First article inspection. - First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I.

TABLE I. First article inspection

| Inspection | Reference |
|---|---|
| Production inspection | 4.4.1 |
| Production control inspection | 4.4.2 |
| Environmental tests | 4.4.3 |

4.3.1 First article inspection report. - Six copies of the first article inspection report shall be submitted to the Naval Electronic Systems Command, Washington, D. C., 20360, via the Government inspector, for review and approval (see 6.3). An advance copy shall be forwarded to NAVELEX Code 0475. The first article inspection report shall indicate specification compliance limits as well as actual performance test results. The supplier shall certify that, on the basis of the report, the equipment does or does not comply with specification requirements.

4.4 Quality conformance inspection. -

4.4.1 Production inspection. - Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden



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defects of material. Production inspection shall consist of the tests approved in accordance with 4.1.3 and shall include the tests specified in table II.

TABLE II. Production inspection

| Inspection | Reference |
|---|---|
| Visual and infrared transmission | 4.4.1.1 |
| Thermal shock test | 4.4.1.2 |
| Infrared filter visual inspection | 4.5.6 |
| Visual security | 4.5.8 |
| Operating test | MIL-E-16400 |
| Surface examination | MIL-E-16400 |

4.4.1.1 Visual and infrared transmission.- The visual and infrared transmission tests shall be performed on samples from each melt of glass to be used in the manufacture of the infrared filters as specified in 4.5.3.

4.4.1.2 Thermal shock test.- The infrared filter shall be tested as a unit in accordance with 4.5.5 and the sampling procedure specified in 4.4.2.

4.4.2 Production control inspection.- Production control inspection, including sampling, shall conform to table III below and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on the equipment and the requirements shall be satisfactorily met prior to release for shipment.

TABLE III. Production control inspection

| Inspection | Inspection Paragraph | AQL Percent |
|---|---|---|
| Weights and dimensions | 4.5.15 | 6.5 |
| Enclosure | 4.5.9 | 6.5 |
| Magnetic test | 4.5.13 | 6.5 |
| Insulation resistance | 4.5.1 | 6.5 |
| Filter optical characteristics | 4.5.10 | 6.5 |
| Thermal strength | 4.5.4 | 6.5 |
| Solar radiation | 4.5.2 | 6.5 |

4.4.2.1 Rejected lots.- If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.2.2 Reinspection of conforming production control test units.- Unless otherwise specified, sample units which have been subjected to and passed production control inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.4.3 Environmental tests.- Environmental tests shall consist of the tests specified in table IV and shall be performed on units which have passed production inspection.



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TABLE IV. Environmental tests

| Inspection | Reference |
|---|---|
| Temperature | 4.5.14 |
| Humidity | 4.5.7 |
| Drop test | 4.5.12 |
| Battery tests | 4.5.11 |
| Salt spray test | MIL-E-16400 |

4.4.3.1 If a sample unit fails the test specified in 4.4.3, the supplier shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the supplier cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.5 Inspection procedures.-

4.5.1 Insulation resistance.- A potential of 500 volts, 60 Hertz, shall be applied continuously for one minute between circuits and between each circuit and the metal external surface of the equipment. Current leakage shall not exceed 5 microamperes. This potential shall not be applied across lamps.

4.5.2 Solar radiation test.- The IR filter shall be subjected to the sunshine test of MIL-STD-810, test method 505, procedure I. Any change in the IR or visual transmission characteristics of the filter as a result of the sunshine test shall be considered as unacceptable.

4.5.3 Visual and infrared transmission test.- Three 2-inch by 2-inch-square samples shall be taken from each melt of glass used in the manufacture of the infrared filters. The samples shall have approximate thicknesses of 25 percent, 50 percent, and 100 percent of the nominal thickness of the IR filter. The samples shall be used for determination of compliance with visual and IR transmission requirements. The visual and infrared transmissions of each sample shall be forwarded to the Naval Electronic Systems Command for approval.

4.5.4 Thermal strength test.- The IR filter shall be held for a period of 24 hours at an ambient temperature of 180°F (82°C) and allowed to cool normally in free air. There shall be no fractures during any part of this test. Plastic filters shall not warp or deform during this test.

4.5.5 Thermal shock test.- The infrared filter shall be heated uniformly to a temperature of 180°F (82°C), then suddenly and completely, or partly, immersed in water at 32°F (0°C) without breaking. Thermal shock tests will be performed upon extra sample filters provided for these tests only. Regardless of condition, filters which have been subjected to these tests shall not be furnished with production equipment or as spares.

4.5.6 Infrared filter visual inspection.- The infrared filter visual inspection shall be conducted to assure dimensional and optical uniformity and to eliminate defects from all filters delivered. The inspection shall include surface examination, and determination of compliance to weight dimensions and visual transmission limitations. The visual transmission test shall be conducted in a darkened area. The filter shall be illuminated from within by a light bright enough to reveal any flaws, seeds, bubbles, cracks, or thin spots in the glass. The filter shall be rotated and inspected from all angles. Any void, bubble or seed exceeding 1/8 of an inch in diameter or a total of defects sufficient to cover 10 percent of the surface of the filter shall be unacceptable.

4.5.7 Humidity test.- The humidity test shall be conducted in accordance with MIL-E-16400. Arcing or excessive leakage to ground shall be cause for rejection until the fault is corrected. Temperature excursions during the temperature and humidity tests shall not exceed the design limits of the storage battery.



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4.5.8 Visual security.- The equipment shall be set up in a completely darkened area and the power turned on. The control panel, switches, and beacons shall be examined for white light leaks. There shall be no white light emitted from any part of the equipment with the exception of void or bubbles allowable under 4.5.7. A uniform red glow transmitted through the filter/dome is natural and is not cause for rejection.

4.5.9 Enclosure tests.- The enclosure tests shall be as specified in MIL-STD-108 for watertight. The equipment shall be submerged in a minimum of 3 feet of water.

4.5.10 Filter optical characteristics test.- Photometric measurements shall be made to determine compliance with 3.3.1 and 3.3.4.2.2.

4.5.11 Battery tests.- The battery tests shall be conducted on the equipment with a fully charged battery in place. Any failure of battery shall be considered an equipment failure. If a battery failure occurs during any test, the cause of failure will be determined and the test repeated with a new battery. If the second battery also fails, tests will be halted until the defect is corrected by the battery manufacturer. Batteries submitted with first article equipment shall be inspected for conformance with 3.3.9. In the event the battery manufacturer is qualified under MIL-B-8565, the inspector may accept an affidavit from the manufacturer in lieu of the applicable tests thereunder. This affidavit shall certify that the batteries furnished on the contract or order conform in all respects to this specification and the applicable requirements of MIL-B-8565.

4.5.12 Drop test.- The equipment shall be dropped from a height of 3 feet onto a horizontal steel plate. It shall be dropped three times, once on the bottom, once on the end, and once on a long side. It shall not be dropped upside down. The equipment shall be capable of normal operation and there shall be no breakage or severe dislocation of parts as a result of the three drops, except that breakage of the lamp shall not be cause for rejection.

4.5.13 Magnetic test.- The magnetic test shall be as specified in MIL-E-16400.

4.5.14 Temperature.- The temperature test shall be performed as specified in MIL-E-16400 for class 2 equipment except that the operating temperature shall be -28° to +60°C.

4.5.15 Weights and dimensions.- Tests for weights and dimensions shall conform to the requirements of 3.3.3.

4.6 Acceptable commercial parts.- For equipment parts such as lamps, cables and storage batteries, the supplier shall certify that the material offered conforms to the following, as applicable:

(a) UL approved.
(b) Conforms to ASA standards.
(c) Complies with specification requirements.

4.7 Preparation for delivery inspection.-

4.7.1 Level A.- Level A packages shall be in accordance with MIL-P-116.

4.7.2 First article inspection.- First article inspection, including the rough handling test, specified in MIL-P-116, is required except where level C packaging and packing is specified. First article inspection will not be required when such a package and pack has previously been inspected and accepted for the same method for a similar item and satisfactory evidence can be furnished to the Government that sets have been prepared identically with previously approved packages and packs.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.1).



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5.1.1 Level A.

5.1.1.1 Cleaning. Cleaning shall be in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Drying shall be in accordance with MIL-P-116.

5.1.1.3 Preservative application. None.

5.1.1.4 Unit packaging. Each infrared transmitting set shall be individually packaged in accordance with MIL-P-116, insuring compliance with the General paragraph under Methods of Preservation (Unit Protection) and the Physical Protection paragraph therein. Cushioning and dunnage shall be used to prevent physical or mechanical damage.

5.1.1.4.1 Major components. The infrared transmitter (beacon) shall be packaged in accordance with Method IIb of MIL-P-116. Desiccant shall conform to MIL-D-3464, Type II or III.

5.1.1.4.2 Other components. The mast, storage battery, and spare lamp shall be packaged in accordance with Method III of MIL-P-116 using containers conforming to PPP-B-636.

5.1.1.4.3 Optical parts. Optical parts shall be preserved and packaged in accordance with MIL-O-16898.

5.1.1.4.4 Equipment repair parts. Equipment repair parts or running spares shall be individually packaged in sealed transparent packages in accordance with MIL-E-17555. These items shall be packaged separately from other items within the pack in containers conforming to PPP-B-636 and identified to the applicable equipment. Repair parts lists shall be provided in accordance with MIL-E-17555.

5.1.1.4.5 Technical manuals. Technical manuals shall be packaged in a waterproof, transparent, heat sealed bag, minimum 4 mil thick, and shall not be placed within the barrier used to package any other items.

5.1.2 Level C. Each complete infrared transmitting set shall be packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Items comprising a complete infrared transmitting set, packaged as specified in 5.1, shall be packed in a fiberboard consolidation container conforming to PPP-B-636, V Grade, Special Use Requirement, for Type 2 Load. In lieu of the closure and waterproofing requirements in the appendix to PPP-B-636, closures and waterproofing shall be accomplished by sealing the center seams, corners, and manufacturer's joint with waterproof tape, 2 inches minimum width, conforming to PPP-T-60, Class 1, or PPP-T-76. Reinforcing shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. Infrared transmitting sets, packaged as specified in 5.1, shall be packed in fiberboard consolidation containers conforming to PPP-B-636, Class weather-resistant. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. Items, packaged as specified in 5.1, shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules, or regulations of other carriers, as applicable to the mode of transportation.

5.3 Unitized loads. When applicable, items packed as specified in 5.2 shall be palletized or containerized in accordance with MIL-STD-147 whenever quantities to one destination equal 40 cubic feet or more. Quantities of packed items too small or impracticable for a unitized load shall be consolidated in shipping



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containers specified in 5.2. Palletized loads shall be strapped to the pallet. Unitized loads shall not be subjected to rough handling tests. When applicable, items for shipboard end destinations shall be unitized by hull number.

5.4 Marking. In addition to any special marking required by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129 (see 6.1). Marking shall include special markings for Method II packs, and dangerous materials markings required by DSAM 4145.3.

5.5 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible. Unless otherwise specified, loose-fill polystyrene shall not be used for packaging or packing. Adhesives shall not be used in the closure of wraps, bags, or boxes.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Levels of preservation-packaging and packing, and applicable marking (see section 5)
(c) Whether first article inspection shall be performed by the supplier or the Government.

6.2 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement. (See 4.3.1)

6.3 Contract data requirements. Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification include, but are not restricted to the following:

a. Contractor's quality assurance system (4.1.1)
b. Tests, test equipment and test procedures (4.1.3)
c. First article inspection report (4.3.1)

6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - EC
(Project 5850-N029)



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DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. *GIVE PARAGRAPH NUMBER AND WORDING.*

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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